

CS & AI • EXPLAINED

What is an **algorithm**?

You already use them every day — without even knowing it.

THE HOOK

Made 2-minute **Maggi** today? You just ran an **algorithm**.

Boil water → add masala → add Maggi → wait 2 min. **Steps, in order.**

That's all an algorithm is — a recipe. Let's make it click. 🍜



TODAY'S QUESTION

What exactly **is** an algorithm —
and why does CS care so **much**
about it?

It's just a **step-by-step** recipe.

...

```
# algorithm: make 1 cup chai
```

```
1. boil 1 cup water
```

```
2. add 1 tsp tea + sugar
```

```
3. add milk, boil 2 min
```

```
4. strain into cup → done ✓
```

- A clear set of steps...
- ...to solve **one** specific problem.
- That's it. That's an algorithm.

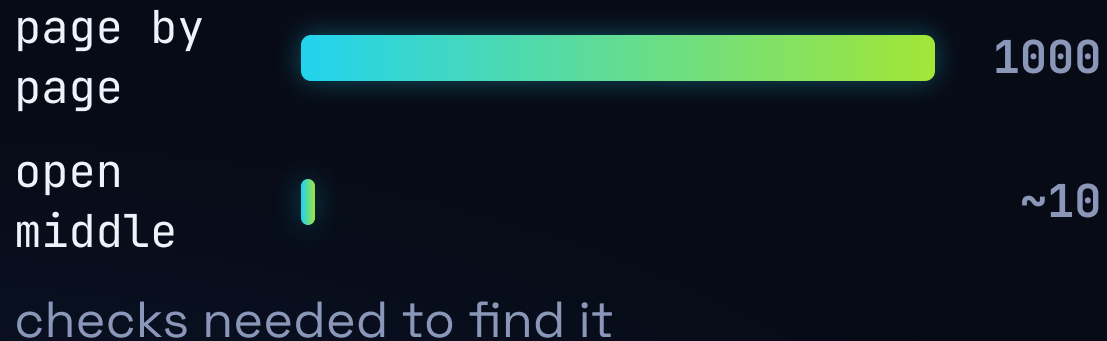
Steps must be **clear, ordered & finite**.

- **Clear:** no "add thoda sa" — how much, exactly?
- **Ordered:** add Maggi *before* boiling water? Disaster. Order matters.
- **Finite:** it must *end* — no running forever.

A computer follows steps **literally**. Be vague, and it breaks.

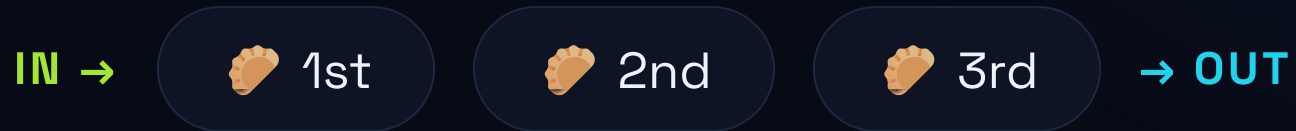
Same task → a smarter algorithm is way faster.

Find a word in a **1000-page** dictionary:



- Page by page = up to 1000 looks.
- Open the middle, go left or right, repeat = ~10 looks.
- Both correct — one is 100× faster.

Simple rules, like a **queue**.



- Samosa line: whoever comes **first**, gets served **first**.
- That's a **queue** — FIFO (First In, First Out).
- Print jobs, ticket counters, WhatsApp messages — all run on a queue.

WHY IT MATTERS

Algorithms run your **whole day**.

- **Google Maps** picks the shortest route home.
- **YouTube** decides which video to show you next.
- **UPI** checks and clears a payment in 2 seconds.

All of it = clever step-by-step algorithms, running millions of times a second.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"Algorithm = scary maths that only coders and AI use."

✓ REALITY

It's just **clear steps** to do a task — you use them daily. Code is only writing those steps precisely enough for a **computer**.

RECAP — 2 THINGS TO REMEMBER

- 1 An algorithm = precise, ordered, finite steps to solve a problem.
- 2 The same task can have a fast or slow algorithm — that's the heart of CS.

Algorithms = **recipes**. Simple. 🍜

▶ Subscribe @KatixoKhojLab

Next question: **How does the internet reach your phone?** 🌐